

CONFIDENTIAL OUTBREAK INVESTIGATION
DAY 1 BRIEF

TO: Industry modeling consultants hired by UVA
FROM: UVA Student Health & Wellness
DATE: February 19, 2026

SITUATION:

An outbreak of a **previously uncharacterized viral illness** has been detected among undergraduate students. Confirmed cases have been reported in multiple residence halls. Laboratory testing confirms a viral pathogen; further classification is ongoing. At this time, only partial case data are available.

Population at risk: 17,900 undergraduate students

CURRENT DATA AVAILABLE

- Daily active case counts for the **first 45 days** of the outbreak
- Case confirmation is based on symptom onset + testing delay (details unavailable)
- No post-peak data are available.

EPIDEMIOLOGICAL FACTS (CONFIRMED)

- Transmission occurs primarily via **respiratory droplets**
- **Pre-symptomatic transmission** has been documented
- No evidence of pre-existing immunity

YOUR TASKS OVER THE NEXT FEW WEEKS

- Estimate the **initial growth rate** from early outbreak data
- **Predict** the spread of the disease and **when cases will peak** using SEIR modeling
- Report best guess for **virus family** based on characteristics of spread
- Recommend a best **intervention strategy to reduce the spread** of the disease while minimizing disruptions to campus life and student learning

CONFIDENTIAL OUTBREAK INVESTIGATION
DAY 2 BRIEF

OUTBREAK UPDATE — UNDERSTANDING EPIDEMIC SPREAD

TO: Industry modeling consultants hired by UVA
FROM: UVA Student Health & Wellness
DATE: February 24, 2026

NEW INFORMATION

Surveillance teams note that early transmission clusters were associated with large indoor gatherings and shared dining facilities. Environmental sampling has detected viral RNA on high-touch surfaces, though viability outside the host appears limited. Routine vaccination records have not demonstrated clear protective effects. Investigations remain ongoing and has refined the clinical timeline.

UPDATED DISEASE PARAMETERS

- **Symptomatic period:** approximately **5-9 days**
 - **Infectious period:** **~2 days** before symptom onset
- **Mild respiratory symptoms** predominate early in illness
- Some students report a **rash** and **low-grade fever**. A minority of cases describe **sore throat** and **fatigue** preceding fever.
- **Gastrointestinal** symptoms have been reported in ~8% of cases.

DATA AVAILABLE

No new data is available at this time. Investigations are ongoing, and new data is expected in the next briefing.

CURRENT TASKS

- Understand the relationship between **disease characteristics** and **epidemic characteristics**
- Estimate the **initial growth rate** from early outbreak data

CONFIDENTIAL OUTBREAK INVESTIGATION
DAY 3 BRIEF

OUTBREAK UPDATE — SEIR MODELING

TO: Industry modeling consultants hired by UVA

FROM: UVA Student Health & Wellness

DATE: February 26, 2026

NEW INFORMATION

Case surveillance now includes daily case counts through **day 70**.

⚠ Note: The outbreak has not yet peaked in the data provided.

UPDATED EPIDEMIOLOGICAL PARAMETERS

- **Incubation period:** approximately **12–18 days**
- Some students are reporting joint pain as a symptom.
- Early clusters of disease have coincided with midterm exam periods, during which library occupancy reached peak capacity.
- Symptom-based testing criteria were broadened after rash presentation was recognized.

CURRENT TASKS

- Choose **best parameters** for SEIR model that fit the data and disease information using a grid search
- **Predict** the spread of the disease and **when cases will peak**
- Plot the model-predicted infections over time

CONFIDENTIAL OUTBREAK INVESTIGATION
DAY 4 BRIEF

OUTBREAK UPDATE — INTERVENTION MODELING

TO: Industry modeling consultants hired by UVA

FROM: UVA Student Health & Wellness

DATE: March 10, 2026

NEW INFORMATION

Case surveillance now includes daily case counts through **day 120**. It appears that the disease spread has peaked, and case counts are now decreasing.

NEW DISEASE EPICENTER

Unfortunately, the disease seems to have spread to Virginia Tech and has been spreading for **70 days**. While the disease spread parameters are the same, VT has a much larger undergraduate population. The VT team requests that you model what would happen if different intervention strategies are adopted at **day 70**.

UPDATED VIROLOGY KNOWLEDGE

- **Genome:** Positive-sense RNA (~10 kb); Virus can be directly translated and does not require transcription
- **Virus has a small size:** 50-80nm

CURRENT TASKS

- Using the information about the peak of infections in the new data, **compare your model-predicted peak to the new data**.
- Recommend a best **intervention strategy to reduce the spread** of the disease while minimizing disruptions to campus life and student learning at Virginia Tech.

CONFIDENTIAL OUTBREAK INVESTIGATION
DAY 5 BRIEF

OUTBREAK UPDATE — VIRUS IDENTIFICATION

TO: Industry modeling consultants hired by UVA

FROM: UVA Student Health & Wellness

DATE: March 12, 2026

NEW INFORMATION

- Prior vaccination records do not appear protective against this strain.
- Symptom surveillance indicates joint pain reports are disproportionately higher among female students, particularly those over age 20.

OPERATIONAL HEALTH ADVISORY (NEW)

Out of an abundance of caution, Student Health recommends that infected individuals avoid pregnant individuals, and consult medical providers if exposure is suspected. No severe maternal illness has been reported.

VT DISEASE EPICENTER: CURRENT TASKS

- The new infections per day are still increasing at this time, suggesting this new epicenter has not yet reached the peak.
- VT's epidemiological modeling team is waiting for your **recommendations for the best intervention** they can take to minimize the spread of this disease while also minimizing disruptions to university life and courses.
- Please detail the **limitations** of your modeling approach and ways our modeling team could improve your model in the future.
- Please share any information you can provide about the **virus**, including family.